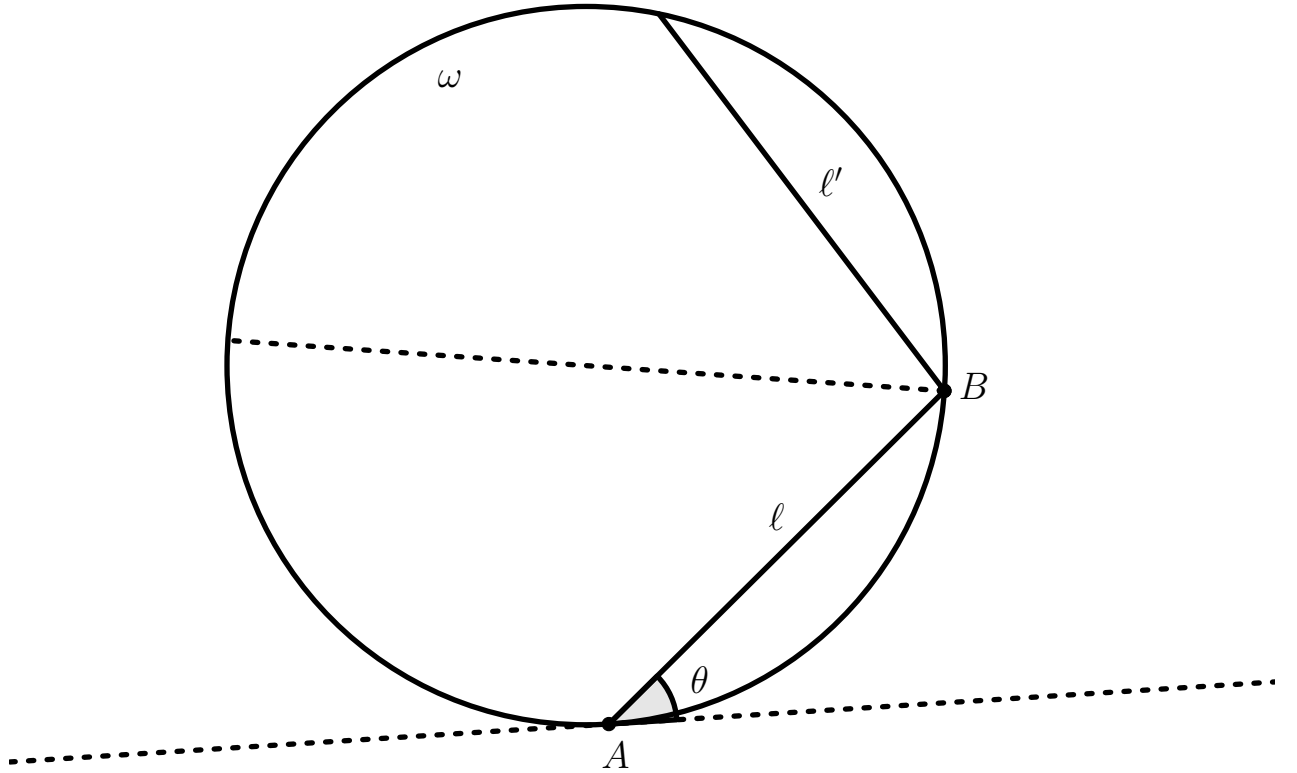


Solutions to Reflective Circles

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Solution 1:

For which values of θ does the laser's path reach point A?

Since the angle made by ℓ and the tangent to point A is θ degrees, arc AB must have a measure of 2θ . ℓ' is a reflection of ℓ , so it is congruent to ℓ . Thus, the arc that ℓ' subtends is congruent to arc AB, which means it has a measure of 2θ as well. Every single bounce made by the laser follows a path that is the result of a sequence of reflections of ℓ , so every single path goes a multiple of 2θ degrees around the circle. Since a circle's circumference covers 360 degrees, the path only ever circles back to point A if there exists an integer n such that $2n\theta$ is a multiple of 360.

If θ is rational, then it can be written as $\frac{p}{q}$, where p and q are relatively prime positive integers. If we let $n = 360q$, then $2n\theta$ is a multiple of 360. If θ is irrational, then $2n\theta$ is also irrational

because the product of a rational and an irrational number is irrational, which means that n only exists for all rational values of θ , so the laser circles back to point A if and only if θ is rational.

Solution 2:

What can we say about the set of points on ω that the laser will never reach?

Since the laser travels 2θ degrees along the circle between bounces, every point it hits must be a multiple of 2θ degrees around the circle from point A . Thus, the set of points that it does not reach is simply the set of points on the circle that are not a multiple of 2θ degrees from point A .

To be more specific, we can look at two cases: either θ is irrational or it is rational.

If θ is irrational, then the laser cannot reach any point that is a rational number of degrees from point A (except for point A itself). We can prove this using contradiction. Assume that there is a point that is $\frac{p}{q}$ degrees from point A that is on the laser's path. Then, $\frac{p}{q} + 360k = 2n\theta$, where k and n are nonnegative integers. However, the left side of the equation is rational, while the right side is irrational, which is a contradiction. Thus, the set of points that the laser cannot reach if θ is irrational includes the set of points a rational number of degrees from point A .

Similarly, we see that if θ is rational, then the laser can never reach a point which is an irrational number of degrees away from point A .